

# Technical Progress Report

IoT-Based Smart Dengue Environmental Risk Monitoring and Early Warning System Using ESP32

Team: **AuraLink** | **Track A** (Embedded IoT)

## Problem

In dense residential lanes of Gampaha and Colombo, Public Health Inspectors and Medical Officers of Health lack continuous, localized visibility of dengue-favourable conditions (warm, humid weather and stagnant water after rain), because surveillance relies on periodic cluster surveys while the Aedes cycle takes only 7-10 days. Sri Lanka recorded 94,666 cases and 72 deaths up to 26 August 2026, about 41% of them in these two districts (NDCU).

## Proposed Solution

An ESP32 samples a DHT22, a rainfall input (potentiometer in simulation) and an ultrasonic water-depth sensor every 5 s and computes a 0-100 risk score:  $\text{Risk} = 100(0.25T_s + 0.20H_s + 0.20R_s + 0.25W_s + 0.10H_i)$ , with Low below 40, Medium 40-69 and High 70 and above. LEDs, buzzer and LCD alert locally; JSON telemetry is published to an MQTT broker.

## Current Progress

- Simulation: Circuit built and running in Wokwi: ESP32, DHT22, potentiometer (rainfall), HC-SR04, 16x2 I2C LCD, three LEDs, buzzer.
- Firmware (Arduino C++): Sensor reading, normalized scores, weighted risk model, Low/Medium/High classification.
- Local alerts: LCD shows live readings and level; LEDs and buzzer respond to the risk level.
- Connectivity: Wi-Fi connects and JSON telemetry is published to the HiveMQ public broker.
- Repository: /firmware, /hardware, /docs and /tests folders set up on GitHub.

## Technology Stack

ESP32 DevKit; DHT22; potentiometer (rain simulation); HC-SR04; 16x2 I2C LCD; LEDs; buzzer. Arduino C++ (PubSubClient, DHT, LiquidCrystal\_I2C); Wokwi; Wi-Fi, MQTT (HiveMQ), JSON; GitHub. Planned: BME280/SHT31, tipping-bucket rain gauge, JSN-SR04T, DS18B20, Node-RED, Telegram bot.

## Testing

Wokwi run, default inputs (T = 24 C, H = 40%, rain 0, water 0): risk score 27.0 (LOW), green LED on, LCD shows Risk:LOW, and "MQTT published" appears every cycle.

Next test: Medium and High scenarios using the potentiometer and ultrasonic distance.

## Limitations

Simulation only; rainfall is a potentiometer, not a real gauge. One node covers one reference container. Weights are a heuristic until calibrated with real case data. The public MQTT broker is not secure. HC-SR04 is not waterproof (JSN-SR04T planned). Stagnation clock, offline buffer and watchdog are not yet implemented.

## Next Step

- (1) Build the physical prototype and replace the potentiometer with a tipping-bucket rain gauge.
- (2) test Medium and High scenarios.
- (3) add the stagnation clock (day 5 / day 7), offline buffer and watchdog.
- (4) move to a private MQTT broker.
- (5) add the Node-RED dashboard and Telegram alerts.
- (6) calibrate the weights with Gampaha / Colombo data.